



Technical
Competency
Assessment



Overview

Thank you so much for taking the time to interview for the Astra Product Engineer position, and congratulations on progressing to the second round! We are so excited to see all the insights that you will mine from this dataset.

Background: You have been tasked with helping a Kingston-based, student-run, residential painting business optimize how it prioritize inbound sales leads. The company generates 200-400 inbound monthly leads through Facebook Ads, lawn signs, and door-to-door canvassing, and spends significant time pursuing low-quality leads that rarely convert. This reduces profitability and limits the team's labour capacity.

The objective is to, by using the data provided, develop a structured way to determine which leads are most likely to convert into profitable jobs. This tool should be usable into the future, rather than creating a one-off solution for the current leads. Additionally, non-technical employees from the sales and marketing teams should be able to interface with this tool in an efficient, simple manner.

You have access to an external API that provides lead data collected from the company's marketing channels. The dataset contains various factors, including property type, neighbourhood, estimated job size, requested timeline, referral source, homeowner status, and other relevant attributes.

Access your data:

You will retrieve 2500 rows of inbound lead data via a Supabase REST API; full endpoint, API key, and pagination instructions are provided on the next page.

Deliverables

Github Repository: A *public* GitHub repository link that contains all of the code used for data analysis and product development. You can code in whatever language you feel most comfortable coding in. Please feel free to use any AI tools (Cursor, Claude Code, etc.) to aid in your coding process.

Recorded Demo: A simple 0-2 minute demo exploring your solution. This does not need to be edited or contain a voice-over - we simply want to see your product in action.

Slide Deck: A presentation summarizing your analysis process, insights, and product. During the in-person interview, you will have 10 minutes to present your findings using this slide deck. The slide deck should contain 4-6 slides (excluding title and transitions) and include the following:

1. Intro
 - Problem Summary
 - Tools used (languages, libraries, etc.)
 - Process Breakdown
2. Analysis
 - Data Exploration and Analysis
 - Product Features
3. Summary
 - Explain how your solution solves the client's problems

Only include Analysis Outputs, do not include code in the presentation

Overview

Analysis Objectives

You have liberty to explore the dataset in whatever way you deem logical. We really want you to be creative and draw insights from the dataset. Given the open-ended scope of this problem, your exploration of the dataset should provide the logical basis for the actions you take when building your solution.

You will be evaluated based on the following criteria:

1. **Methodology:** The soundness of your analysis methodology and techniques employed.
2. **Insightfulness:** The depth and relevance of the insights obtained from the data.
3. **Problem-fit:** The degree to which your product solves the client's problem in an accessible and maintainable manner
4. **Communication:** Clarity and coherence of your presentation and code.

Submission Instructions

When you are finished the deliverable:

- Save your Slide Deck as a PDF as:
"Your Name – Astra TCA Presentation"
- Submit your slide deck, alongside the link to your *public* Github repository in the following form:
<https://forms.gle/BJFesInBUshsBpxHA>
- Sign up for a second-round interview slot using the following link:
<https://calendly.com/creosolutions-smit/h/creo-solutions-astra-2026-2027-second-round>

API Access

Base URL:

<https://dqcqexieqlfqylqwogsj.supabase.co>

Table Endpoint:

GET /rest/v1/inbound_leads

API Key (include this in the request header):
apikey:sb_publishable_iFgXIEVP5UqmrZx6l1nVgw_6WD5CPpy

Pagination:

The API enforces a maximum number of rows returned per request. To fetch the full dataset, you must paginate using limit and offset.

Additional Information

We encourage you to explore the dataset and develop a well-reasoned, complete product backed by clear, logical analysis. Be sure to document how you arrived at your conclusions.

We also encourage you to be creative in the manner that you allow the sales and marketing team to interact with your tool.

The use of AI tools is encouraged, but be prepared to answer questions about your code, and the design decisions you made. Simply put, you should understand *all of the code that you submit*.

If you have any questions or need clarification on the assignment, feel free to reach out to:

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